

HNSS18 · Eye & Ear Development — Timeline

Human ages approximate · **EXAM** = flagged in class as testable · Origin key: **Neuroectoderm** **Surface ectoderm** **Neural crest** **Endoderm (pouch 1)** **Arch 1**
Arch 2 **Mesenchyme**

HoyaDocs
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The Eye				
AGE	STAGE / STRUCTURE	WHAT HAPPENS	ORIGIN	CLINICAL LINK
~Day 17	Eye fields	Cells in the anterior neural plate (prosencephalon), in line with the diencephalon , form the eye fields EXAM ; they express PAX6 . SHH from the anterior ventral midline represses PAX6 in the midline → splits one field into two	Neuroectoderm	No SHH → fields converge → cyclopia
Days 21–25	Optic grooves (sulci)	Rapid growth of the prosencephalon carries the eye fields forward; cells invaginate → optic grooves . Cranial neural folds approach the midline; groove neuroectoderm nears the surface ectoderm	Neuroectoderm	—
Day 28	Optic vesicles + stalks	Optic grooves form the optic vesicles and optic stalks . Vesicle contacts surface ectoderm → induces the lens placode (Pax6 + Sox2 make the ectoderm competent to respond to FGF/BMP)	Neuroectoderm	—
Days 30–33	Lens placode → lens pit	Surface ectoderm thickens → lens placode → invaginates → lens pit . Reciprocal signal makes the optic vesicle invaginate	Surface ectoderm	—
Day 36	Lens vesicle + optic cup	Lens vesicle pinches off from the surface ectoderm. Optic vesicle invaginates → bilayered optic cup , still connected to the forebrain by the optic stalk . Lens induces the cornea EXAM	Surface ectoderm Neuroectoderm	—
Week 6	Choroid fissure + hyaloid artery	Optic cup and stalk invaginate inferiorly → choroid fissure ; the hyaloid artery runs through it to the lens. Later: proximal part → central artery of the retina ; distal part degenerates → hyaloid canal in the vitreous	Neuroectoderm Mesenchyme	Fissure fails to close → coloboma (inferior "keyhole" iris)
Week 7	Lens · cornea · retina · anterior chamber	Posterior lens fibers elongate and fill the lens cavity ; cornea begins to differentiate; pigmented layer thins while the neural retina thickens (only the neural retina has photoreceptors EXAM); anterior chamber opens between the lens/iridopupillary membrane and the cornea	Surface ectoderm Neuroectoderm	—
Week 8	Lens growth · cornea layers · eyelids begin	Lens grows by secondary fibers added at the equator (Pax6, Sox2, Maf). Cornea = epithelium (surface ectoderm) + stroma & endothelium (neural crest). Eyelids (surface ectoderm + NC) begin to form by the end of the embryonic period	Surface ectoderm Neural crest	—
Weeks 8–9	Iris	Anterior rim of the optic cup → iris epithelium + sphincter & dilator pupillae ; neural crest mesenchyme → iris stroma	Neuroectoderm Neural crest	PAX6 mutation → aniridia STEP 1
~Week 10	Sclera + choroid	Mesenchyme (neural crest) around the optic cup condenses → choroid (vascular, inner) and sclera (fibrous, outer)	Neural crest	—
Week 15	Ciliary body	Folding of the two optic-cup epithelial layers → ciliary processes (ciliary epithelium); neural crest → ciliary muscle + stroma . Processes secrete aqueous humor ; suspensory ligaments → accommodation	Neuroectoderm Neural crest	—
2nd → 3rd trimester	Eyelids fuse, then reopen	Eyelids fuse at the start of the 2nd trimester and reopen at the start of the 3rd trimester	Surface ectoderm	—
Before birth	Pupillary membrane regresses	The pupillary membrane normally regresses	Mesenchyme	Persists → strands across pupil; if it blocks the visual axis → deprivation amblyopia

Sequence hook: Fields → Grooves → Vesicles → Cup (+ lens) → Fissure → fill in the front (cornea, iris, ciliary body) → lids fuse and reopen.

★ **USMLE high-yield:** Inner cup → **neural retina**; outer cup → **RPE**. Lens + corneal epithelium = surface ectoderm; corneal stroma/endothelium, iris stroma, ciliary muscle = neural crest. **Iris sphincter/dilator = neuroectoderm**. Coloboma = failed choroid fissure closure. SHH loss → cyclopia (holoprosencephaly spectrum).

The Ear

AGE	STAGE / STRUCTURE	WHAT HAPPENS	ORIGIN	EAR PART / CLINICAL LINK
~Day 22	Otic placode	Thickened surface ectoderm beside the hindbrain (rhombencephalon) becomes the otic placode — the future inner-ear epithelium. The inner ear develops first	Surface ectoderm	Inner ear
Days 26–28	Otic pit	Placodal ectoderm invaginates → otic pit , located dorsal to the 2nd pharyngeal cleft . Meanwhile, 1st pouch endoderm begins to line the future middle ear cavity	Surface ectoderm Endoderm	Inner + middle ear
Day 29	Otic vesicle	Otic pit deepens and pinches off from the surface → otic vesicle (otocyst)	Surface ectoderm	Inner ear
Day 30	Statoacoustic ganglion	Neuroblasts delaminate from the otic epithelium and migrate to form the statoacoustic (vestibuloacoustic) ganglion between the vesicle and neural tube → later vestibular + spiral ganglia	Surface ectoderm	CN VIII
Day 36	Otic vesicle divides	Dorsal → endolymphatic duct + sac · Utricular → utricle + semicircular ducts · Saccular → saccule + cochlear duct	Surface ectoderm	Inner ear (membranous labyrinth)
Week 5	Auricular hillocks	Six auricular hillocks appear around cleft 1: 1–3 on arch 1, 4–6 on arch 2 . External ears start caudal to the lower jaw. Hillock 1 → tragus , 6 → antitragus	Arch 1 Arch 2	External ear
Weeks 5–6	Tubotympanic recess HNSS03	1st pouch → tubotympanic recess → tympatric cavity (distal) + auditory tube (proximal). 1st cleft → external auditory meatus; cleft 1 + pouch 1 + mesenchyme → tympatric membrane	Endoderm Surface ectoderm	Middle + external ear
Weeks 6–9	Auricle takes shape	Hillocks fuse into the auricle (the lobule is not from the hillocks). Lower-jaw growth moves the ears higher and more vertical	Arch 1 Arch 2	Abnormal hillocks or poor jaw growth → low-set / malformed ears → search for other pharyngeal-arch defects EXAM
Week 8 HNSS03	Ossicles	Meckel's cartilage disappears except its dorsal ends → malleus + incus ; Reichert's cartilage → stapes	Arch 1 Arch 2	Middle ear ; Treacher Collins → conductive loss STEP 1
Week 9	Cochlear duct → organ of Corti	An outgrowth of the sacculle forms the cochlear duct ; its specialized epithelial cells differentiate into the organ of Corti (hearing receptors)	Surface ectoderm	Inner ear ; congenital rubella / CMV → sensorineural loss STEP 1

Sequence hook: Placode → Pit → Vesicle → Ganglion → Divide (dorsal / utricular / saccular) → Corti — while the hillocks build the auricle outside. “**Inner = placode, Middle = pouch, Outer = cleft.**”

★ **USMLE high-yield:** Inner ear (membranous labyrinth) = **surface ectoderm** (otic placode). **Pouch 1** → tympanic cavity + auditory tube; **cleft 1** → external auditory meatus; tympanic membrane = all three germ layers. **Malleus/incus = arch 1, stapes = arch 2**. Utricle → semicircular ducts (balance); saccule → cochlear duct (hearing).